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RESEARCH ARTICLE

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A view-tree method to compute viewsheds from digital elevation models

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ABSTRACT

Viewshed computation is a central component for visibility analysis. The majority of existing viewshed computation methods use regular square grid digital elevation models (DEMs) and have limitations in considering spatial relationships among observers, targets and line-of-sight (LOS). Therefore, considering that the visibility is dominated by LOS between observers and targets, this study proposes a method to compute viewsheds using a new type of tree structure, the 'view-tree', that can efficiently extract LOS from DEMs for viewshed computation. The proposed method first constructs a view-tree using cells in regular square grid DEM according to the spatial occlusion relationships among cells. Then, the method traverses the view-tree to judge the visibility of each tree node and derives the viewshed. The results show that the view-tree method is 40% faster than the traditional R2 method. The view-tree method also reduces the error and omission rates by approximately 60% compared with the popular XDraw method.

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Digital terrain analysis; DEM; view-tree; viewshed; visibility analysis

1. Introduction

A viewshed is the ground range that can be observed from one or more locations (Zhou and Liu 2010, Tang *et al.* 2011). Viewshed computation is an important component of visibility analysis and digital terrain analysis (Song 2013). Supported by DEMs, viewshed computation is also a classical geographical problem that integrates many principles of geographical information science and methods in geospatial analysis (Pan et al. 2020, Li 2018). The computation of viewshed has applications across a wide range of fields, for example, telecommunications (Sawada *et al.* 2006), military (Andrade *et al.* 2011), site planning (Yu *et al.* 2016, Wang and Dou 2020) and prospecting (Zhang *et al.* 2003). In the context of high-speed coverage of geographical information, viewshed computation also assists in solving problems such as 3-D landscape

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evaluation (Palmer 2019), surveillance network design (Han et al. 2019) and traffic route planning (Lee and Stucky 1998, Bartie and Kumler 2010).

The earliest method to compute viewsheds involves mapping the terrain profile by contour maps (Song 2013). An increasing number of viewshed computation methods have been proposed by scholars with the emergence and development of related theories in geographical information science (Song 2013). Compared with the traditional viewshed computation methods based on contours, current methods are fast, efficient and widely used. Additionally, the idea of parallel viewshed computation, which can significantly improve the computational efficiency of current serial algorithms, has been proposed (Zhao *et al.* 2013, Feng *et al.* 2015, Song *et al.* 2016, Xu *et al.* 2018).

The existing methods for viewshed computation are generally classified into multiplexed and nonmultiplexed methods according to the reusability of computed locations (Song 2013). Nonmultiplexed methods, such as R3 (Shapira 1990), traverse each location to be computed in the region and use the LOS between the current location and the observer to compute the viewshed. The time complexity of these methods is $O(N^3)$. Nonmultiplexed methods compute the visibility between all locations and the observer. Thus, they possess a relatively high time complexity and have not been widely used. However, the results obtained by nonmultiplexed methods possess the highest precision in principle, and they are therefore widely used for precision evaluations and comparisons with other methods.

As the spatial resolution of elevation data becomes ever finer, efficient methods are increasingly important for meeting the requirements for processing massive amounts of data. Some scholars use efficient algorithms such as dynamic programming and have proposed multiplexed methods that store the visibility of computed locations for computing and updating the visibility of the rest (Franklin and Ray 1994, Zhu *et al.* 2019, Wang *et al.* 2000, Wu *et al.* 2021). R2 and XDraw are two representative multiplexed methods (Song 2013). R2 also computes viewsheds based on LOS and is similar to R3. When the LOS of the peripheral locations is computed, the visibility information of the interior locations can be simultaneously obtained. Therefore, R2 can obtain the entire viewshed by calculating only the LOS of the peripheral locations. The time complexity of the method decreases to $O(N^2)$ compared with R3. XDraw uses the visibility of the computed locations in the inside layer to calculate and update the visibility of the unvisited locations in the outside layer. The method traverses the DEM from the inside layer to the outside layer. Its time complexity is also $O(N^2)$.

Although the time complexity of R2 and XDraw are both $O(N^2)$, they still differ in efficiency and precision because their computational strategies are quite different (Lu *et al.* 2006). On the one hand, although R2 theoretically reduces the traversal time for most locations, it still involves the calculation of LOS and the search for adjacent locations. XDraw only takes the window extent of the current location into consideration during traversal and does not involve the calculation of the whole LOS. As a result, XDraw is superior to R2 in terms of efficiency. On the other hand, R2 has a higher similarity to R3 in computing strategies and therefore produces approximately the same results with high precision. As a comparison, the precision of XDraw is relatively lower than that of R2 because of the distortion of LOS. Moreover, XDraw is only

applicable to smooth and continuous terrain. Therefore, R2 is superior to XDraw in terms of precision.

Some subsequent methods mostly directly use or improve the ideas of R2 and XDraw. For example, the line-rasterization method applies the Bresenham algorithm to simplify LOS, greatly improving the efficiency of R2 by reducing the calculation of searching adjacent locations in the original R2 method (Bresenham 1965, Kaucic and Zalik 2002). The reference plane method replaces the interpolation lines in XDraw with the interpolation planes, which improves the precision of the original XDraw method (Wang *et al.* 2000). The backtrack method and the HiXDraw method separately solve the problem of chunk distortion of LOS in the original XDraw method to some extent (Izraelevitz 2003, Zhu *et al.* 2019). However, studies on combining the advantages of R2 and XDraw are relatively scarce, and only a few subsequent innovative designs in the overall computational strategies have been proposed thus far.

For existing methods for computing viewsheds, the universal contradiction between precision and efficiency is rooted in the trade-off problem of LOS calculation (Wu *et al.* 2021). The quality of LOS is the basic unit and core issue of viewshed computation. On the one hand, the whole LOS calculation can effectively prevent the distortion of visibility of each location, and therefore the precision of the results is guaranteed. On the other hand, the calculation requires a large amount of interpolation and spatial selection of the same or similar locations, so repeated calculation will take a long time, reducing the efficiency of the method. The reason why some existing viewshed computation methods based on local LOS can reduce the number of calculations is that they replace the complete LOS calculation with the visibility of neighbors using dynamic programming. In the present study, a new viewshed computation method is proposed based on a special tree data structure – the view-tree. The regular square grid DEM is converted into a view-tree by judging the spatial occlusion relationship on the two-dimensional plane. Then, we compute the viewshed from the view-tree with traversal algorithms. In the view-tree method, the LOS between the target location and the observation location is constructed indirectly and stored in the tree according to the spatial occlusion relationship without directly performing the complete LOS calculation.

2. Method principle

2.1. Visibility relationship

Because LOS propagates along a straight line, there ought to be a strict occlusion relationship between the front and rear locations on the same LOS. In a regular square grid DEM, elevation values are arranged in discrete cells with specific locations. Thus, the strict occlusion relationship obtained by the traditional cover and intersection algorithms is not applicable for such data (Zhou and Liu 2010).

Some scholars have proposed interpolation to calculate LOS using grid DEMs. A large number of interpolation methods have therefore been designed, such as Janus, Dyntacs and ModSAF (Zhou and Liu 2010). Such methods require high continuity of terrain (Li *et al.* 2020, 2022, Xiong *et al.* 2021), thereby complicating their application to all types of terrain conditions. Furthermore, the large interpolation calculation is

time-consuming and does not make good use of the simplicity of the calculation and indexing in matrix operations.

In this paper, we propose the spatial concept of the view-block relationship to more efficiently express the occlusion relationship between cells in a regular square grid DEM. The view-block relationship describes the interaction and influence between two or more lines starting from the same observer in the plane when they are close to each other. By calculating the view-block relationship, it is possible to determine which objects may cause occlusions to the observer. The view-block relationship on the two-dimensional plane is defined as follows.

Three points *A*, *B* and *C* are given on a two-dimensional plane. The parameter *R* represents the radius of the buffer zone of the LOS. *L*(*A*, *B*) is the connection of *A* and *B*, as the LOS between *A* and *B*. *Buf*(*A*, *B*, *R*) is the buffer zone with radius *R* of *L*(*A*, *B*). If *C* falls within *Buf*(*A*, *B*, *R*), then *C* generates view-block between *A* and *B*; if *C* does not fall within *Buf*(*A*, *B*, *R*), then *C* does not generate view-block between *A* and *B* (Figure 1).

The view-block relationship can be calculated and judged using vectors. The coordinates of *A* (X_A, Y_A), *B* (X_B, Y_B) and *C* (X_C, Y_C) are given to determine whether *C* generates view-block between *A* and *B* (Figure 2).

Let *h* be the distance from *C* to *AB*, angle α be the angle between *AC* and *AB*, and angle β be the angle between *BC* and *BA*. *C* generating view-block between *A* and *B* is numerically equivalent to the formations below.

$$h \leq R \tag{1}$$

$$\alpha < \frac{\pi}{2} \tag{2}$$

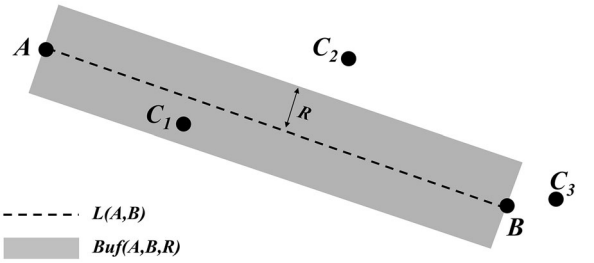


Figure 1. View-block relationship. *L*(*A*, *B*) and its buffer zone *Buf*(*A*, *B*, *R*) with radius *R* are shown. C_1 falls within *Buf*(*A*, *B*, *R*), generating view-block between *A* and *B*. Because C_2 is too far away from *L*(*A*, *B*), and C_3 is not located between *A* and *B*, they are unable to generate view-block between *A* and *B*.

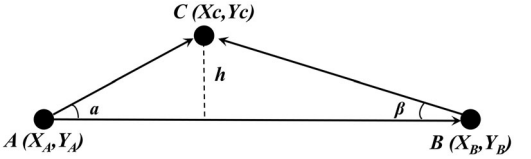


Figure 2. View-block calculation using vectors.

$$\beta < \frac{\pi}{2} \tag{3}$$

By using vectors, these formations can be written as follows:

$$\overrightarrow{AC} \times \overrightarrow{AB}\,|\overrightarrow{AB}| \le R \tag{4}$$

$$\frac{\overrightarrow{AC} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}| \times |\overrightarrow{AC}|} > 0 \tag{5}$$

$$\frac{\overrightarrow{BC} \cdot \overrightarrow{BA}}{|\overrightarrow{BA}| \times |\overrightarrow{BC}|} > 0 \tag{6}$$

By using coordinates, these formations can be written as follows:

$$\frac{(X_C - X_A) \times (Y_B - Y_A) - (Y_C - Y_A) \times (X_B - X_A)}{\sqrt{(X_B - X_A)^2 + (Y_B - Y_A)^2}} \le R \tag{7}$$

$$\frac{(X_C - X_A) \times (X_B - X_A) + (Y_C - Y_A) \times (Y_B - Y_A)}{\sqrt{(X_B - X_A)^2 + (Y_B - Y_A)^2} \times \sqrt{(X_C - X_A)^2 + (Y_C - Y_A)^2}} > 0 \tag{8}$$

$$\frac{(X_C - X_B) \times (X_A - X_B) + (Y_C - Y_B) \times (Y_A - Y_B)}{\sqrt{(X_B - X_A)^2 + (Y_B - Y_A)^2} \times \sqrt{(X_C - X_B)^2 + (Y_C - Y_B)^2}} > 0 \tag{9}$$

The proposed spatial relationship of the view-block is vital for the computation of viewshed from regular square grid DEMs. The estimated view-block relationship may have some deviation in the real world but is an effective and efficient method of abstracting and expressing the strict occlusion relationship between cells in regular square grid DEM (Figure 3). Additionally, because the buffer radius R can be artificially set, this relationship can be applied to various resolutions of DEM and meet different needs of viewshed computation under a range of conditions.

2.2. View-tree data structure

A view-tree is a tree data structure based on the view-block relationship among cells in regular square grid DEM. In one view-tree, each node corresponds to one DEM cell. Given the observation cell *A*, the view-tree is defined as follows:

In the case where the observation cell *A* is given, descriptions like '*C* generates view-block between *A* and *B*' in the following are all abbreviated as '*C* generates view-block for *B*' or '*B* is view-blocked by *C*'.

1. The root node is the observation cell *A*.
2. The root node has eight child nodes corresponding to eight cells in its neighbors.
3. For any other node, the parent node is the cell in its neighbors that can generate view-block for itself. When a node is view-blocked by more than one cell in its neighbors, choose the cell with the smallest *R* as its parent node.

Figure 4 illustrates the construction process of the view-tree with a small part of a DEM where *A* is the root node and *B* is its child node. In *B*'s neighbors, *D* and *C* are

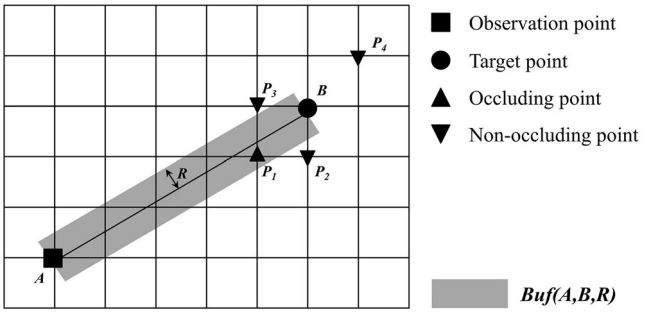


Figure 3. View-block relationship in regular square grid DEM. The observation point *A*, the target point *B*, and several relevant points are shown. Given the buffer radius *R*, P_1 generates view-block, P_2, P_3 and P_4 do not generate a view-block.

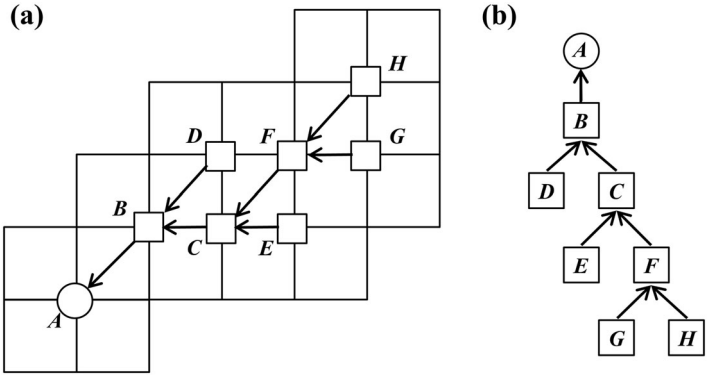


Figure 4. Process and result of constructing view-tree: (a) view-block relationship between cells in DEM; (b) view-tree corresponding to DEM.

view-blocked by *B* so *B* has child nodes *C* and *D*. In *C*'s neighbors, there are *E* and *F* view-blocked by *C* so *C* has child nodes *E* and *F*. In *F*'s neighbors, *H* and *G* are view-blocked by *F* so *F* has child nodes *H* and *G*. Because there is no point in the neighbors of *D*, *E*, *G* and *H* that can be view-blocked, these points are directly regarded as leaf nodes, i.e. they do not have child nodes.

2.3. Viewshed computation

A view-tree method is proposed to compute the viewshed of a given observation location in regular square grid DEM. Based on the construction of the view-tree, we can use depth-first traversal or breadth-first traversal of the entire tree, which is equivalent to the traversal of all cells in the regular square grid DEM. In the process of traversal, the actual visibility slope of the current cell is calculated and used for comparison with the minimum visibility slope to determine the visibility of the current cell. When the traversal is finished, all nodes of the tree will record their minimum visibility slope, and we can therefore obtain the viewshed in the meantime.

Based on the existing relevant theories and research, the minimum visibility elevation of a cell is defined as the minimum elevation that can make the cell visible, and

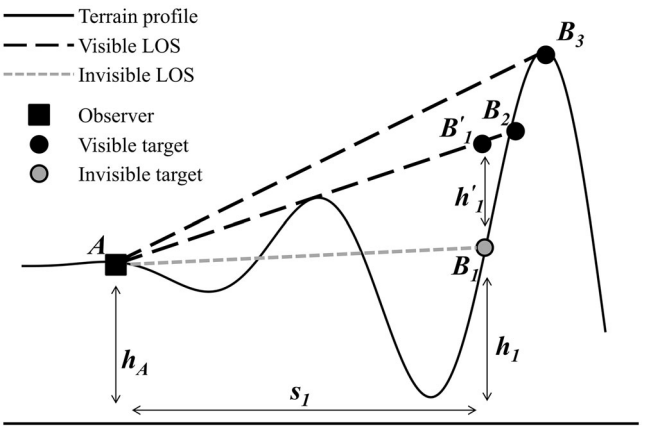


Figure 5. An instance of the minimum visible elevation and minimum visible slope. For B_1, the elevation is h_1, the minimum visible elevation is $h'_1 + h_1$, and the minimum visible slope is $(h'_1 + h_1 - h_A)/s_1$.

Table 1. Explanations of parameters in view-tree method.

Parameter	Data Type	Annotation
AS	Double	The actual slope of LOS connecting target and observer
MVS	Double	The minimum visible slope of the target
O	Coordinate	Coordinate of observer
T	Coordinate	Coordinate of target
N	Coordinate	Coordinate of neighbor
DEM	Array	Elevation of each cell
VS	Array	Viewshed of the area (visibility of each cell)
MVSA	Array	The minimum visible slope of each cell
VA	Array	Traversal state of each cell

the minimum visibility slope is the slope of the LOS between the minimum visibility elevation and the observation location. When the elevation of a cell is greater than or equal to the minimum visibility elevation of the cell, the cell is marked visible (Figure 5). As shown in Figure 5, A is the observation point and B_1, B_2 and B_3 are the target points. For B_1, in its LOS to *A*, there is an intersection part with the terrain profile. So B_1 is marked invisible. For B_2, its LOS to *A* is exactly tangent to the terrain profile. So B_2 is marked visible, and the minimum visibility elevation of B_2 is exactly its own elevation. For B_3, its LOS to *A* does not intersect with the terrain profile. So B_3 is marked visible.

The specific process of the method and explanations of parameters (Table 1) are as follows:

Algorithm: View-tree method

INPUT: (1) digital elevation model of the area *DEM*
(2) observer *O*
OUTPUT: (1) viewshed of the area *VS*
(2) minimum visible slope of the area *MVSA*
1. *VS* ← an empty array of the same shape as *DEM*
2. *MVSA* ← an empty array of the same shape as *DEM*

```
3. VA ← an empty array of the same shape as DEM
4. FUNCTION DFS (target T, minimum visible slope MVS)
5.          IF T is already visited
6.             RETURN
7.          ELSE
8.             AS ← actual slope of LOS connecting T and O
9.          IF AS > MVS
10.            T in VS ← visible
11.            MVS ← AS
12.      ELSE
13.            T in VS ← invisible
14.      T in MVSA ← MVS
15.      T in VA ← visited
16.      FOR each neighbor N in T's neighbors
17.            IF N is view-blocked by T
18.               DFS (N, MVS)
19.            RETURN
20. O in VS ← visible
21. FOR each neighbor N in O's neighbors
22.            DFS (N, -99999999)
23. RETURN VS, MVSA
```

It should be noted that for a different buffer radius R in judging the view-block relationship, the computed viewshed may be different. When R is relatively large, a view-block is more likely to occur between cells and each node may obtain more information from its parent's siblings. When R is small, it is relatively difficult for a view-block to occur, and each node may have fewer child nodes.

When R is too small, it becomes difficult for a view-block to occur. It is possible that one cell is not view-blocked by any other cell in its neighbors, because of which the cell cannot appear in the view-tree and therefore cannot be traversed. To ensure the integrity of the traversal, the buffer radius R has a minimum threshold when set.

When R is too large, view-blocks will become common, and it is possible that a cell is view-blocked by multiple cells in its neighbors at the same time. That means the cell's visibility is affected not only by its parent but also by its parent's siblings. When this happens, the final decision-making mode can be set by the user. According to different decision-making modes, the view-tree method can be reformed into Quick-view-tree, Avg-view-tree, Min-view-tree and Max-view-tree. In this case, depth-first traversal is not recommended because it cannot guarantee that every node is visited after its parent's siblings. In the Quick-view-tree the visibility is determined only by the parent of the current node, and the subsequent traversal to the node will be directly returned without calculation. The Avg-, Min- and Max-view-trees, respectively, use the average (or interpolated), minimum or maximum *MVS* of the parent and parent's siblings as the *MVS* of the current node to determine the visibility.

The time complexity of the view-tree method can reach $O(N^2)$, where N is the number of horizontal or vertical DEM cells. For space complexity, it is necessary to introduce stacks or queues equivalent to the size of LOS to store the minimum visible slope and to maintain arrays equivalent to the size of DEM to record the traversal states of these nodes, so the space complexity is $O(N^2)$. Further, because of the fixed location of the observation cell, information such as the modulus of vector and the actual visible slope can be calculated in advance to reduce the calculation time and improve the reusability of the existing results. This choice requires a space complexity of $O(N^2)$ but can greatly improve computational efficiency.

3. Case study region and method implementation

3.1. Case study region

The selected experimental region is a typical loess hilly-gully region with a regional average altitude of 1035 m in Suide County, Yulin City, Shanxi Province, China. The resampled resolution of the experimental data is 2 m.

The experimental region is located in the Loess Plateau hilly-gully-landform-type area in Northern Shanxi. The overall terrain is high in the east and low in the west. The geomorphic landscape in this region is usually characterized by hills and gullies, where gravity erosion on the surface such as slide and collapse is very active and strong. Erosion gullies are widely distributed along the east-west direction and develop into a complete watershed geomorphic unit. The distribution of terrain feature points and lines in the watershed is relatively natural and dispersed, and the fractal characteristics of the terrain are obvious. The characteristics mentioned above make the selected region suitable for our experiments.

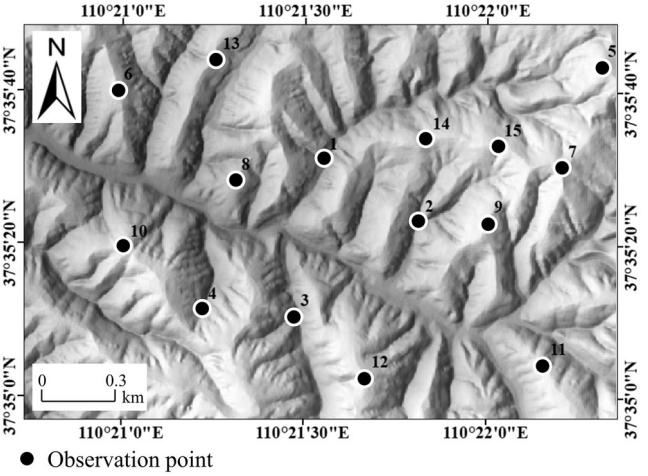


Figure 6. Experimental region and spatial distribution of the chosen observation points.

Table 2. Information of selected terrain feature points.

Point ID	Point Type	Elevation (m)	Longitude	Latitude
1	Runoff node	980	110°21′32″	37°35′31″
2	Runoff node	994	110°21′48″	37°35′22″
3	Runoff node	1023	110°21′28″	37°35′10″
4	Runoff node	1010	110°21′13″	37°35′11″
5	Mountain peak	1134	110°22′19″	37°35′43″
6	Mountain peak	1060	110°20′58″	37°35′39″
7	Mountain peak	1135	110°22′12″	37°35′29″
8	Mountain peak	1030	110°21′18″	37°35′27″
9	Mountain peak	1099	110°22′01″	37°35′22″
10	Mountain peak	1073	110°20′59″	37°35′19″
11	Mountain peak	1078	110°22′09″	37°35′04″
12	Mountain peak	1107	110°21′40″	37°35′02″
13	Saddle point	1046	110°21′15″	37°35′43″
14	Saddle point	1062	110°21′50″	37°35′33″
15	Saddle point	1098	110°22′02″	37°35′32″

3.2. View-tree method implementation

The performance of the view-tree method was tested based on the experimental region described above. For the experiment, we selected 15 typical terrain feature points in the region (Figure 6), including mountain peaks, runoff nodes and saddle points (Table 2), and calculated the viewsheds of each of them. For comparison, we also implemented R3, R2 and XDraw. All calculations for this evaluation were executed using the same hardware configuration.

The specific experiments were divided into three parts: (1) We compared the precision of view-tree method, R2 and XDraw using the error rate η_{er} and omission rate η_{or}. (2) We compared the efficiency of the view-tree method, R2, and XDraw using the time ratio η_{time}. Then we analyzed how the time ratio η_{time} of these methods would vary with the radius of the experimental region individually. (3) We tested the sensitivity of parameter R in view-tree method. Then we analyzed the influence of the parameter on the method's performance.

We evaluated the precision and efficiency of viewshed results computed by view-tree method, R2 and XDraw, with reference to results from the R3 method. To measure the precision of these methods, we use the error rate η_{er} and the omission rate η_{or}. The error rate η_{er} refers to the ratio that is not visible in R3 but is misjudged as visible. The omission rate η_{or} refers to the ratio that is visible in R3 but is misjudged as invisible. Either the error ratio η_{er} or the omission ratio η_{or} indicates good performance with a small value. Let the method to be analyzed be X:

$$\eta_{er}(X) = \frac{N_{R3=invisible \cap X=visible}}{N_{R3=invisible}} \tag{10}$$

$$\eta_{or}(X) = \frac{N_{R3=visible \cap X=invisible}}{N_{R3=visible}} \tag{11}$$

$\eta_{er}(X)$ is the error rate of X, and $\eta_{or}(X)$ is the omission rate of X.

The time ratio η_{time} is used to measure the efficiency of these methods. The absolute time consumed by computation is seriously affected by the difference in operating environment and hardware configuration. Therefore, to improve the intuitiveness

of our experiments, the time ratio η_{time} is used as a relative efficiency measurement. This is the ratio of the operating time of the method compared with that of R3.

Additionally, to analyze the influence of region size on the efficiency of these methods, 10 groups of DEMs, ranging from 50 cells to 500 cells in radius, were used as experimental data to test the operating time and its variation trend of each method under different region sizes.

Finally, a contrast experiment was run to determine the influence of parameter R on the results of the view-tree method. As mentioned in Section 3.2, it is necessary to set the parameter R, the buffer radius of LOS, in advance to determine the view-block relationship on the two-dimensional plane. The view-block relationship would differ for different R values. Then, the view-tree and viewshed would also have some distinctive characteristics, together with the operating efficiencies. Setting the cell size of DEM as L (DEM with $2\,\text{m} \times 2\,\text{m}$ resolution is used as the experimental data, so $L = 2\,\text{m}$), $0.40\,L$ to $0.80\,L$ as the variation interval, and $0.05\,L$ as the step size of R, the viewshed of the experimental data with different values of R were calculated to explore the effect of parameter R on the operating results and on the operating efficiency of view-tree method.

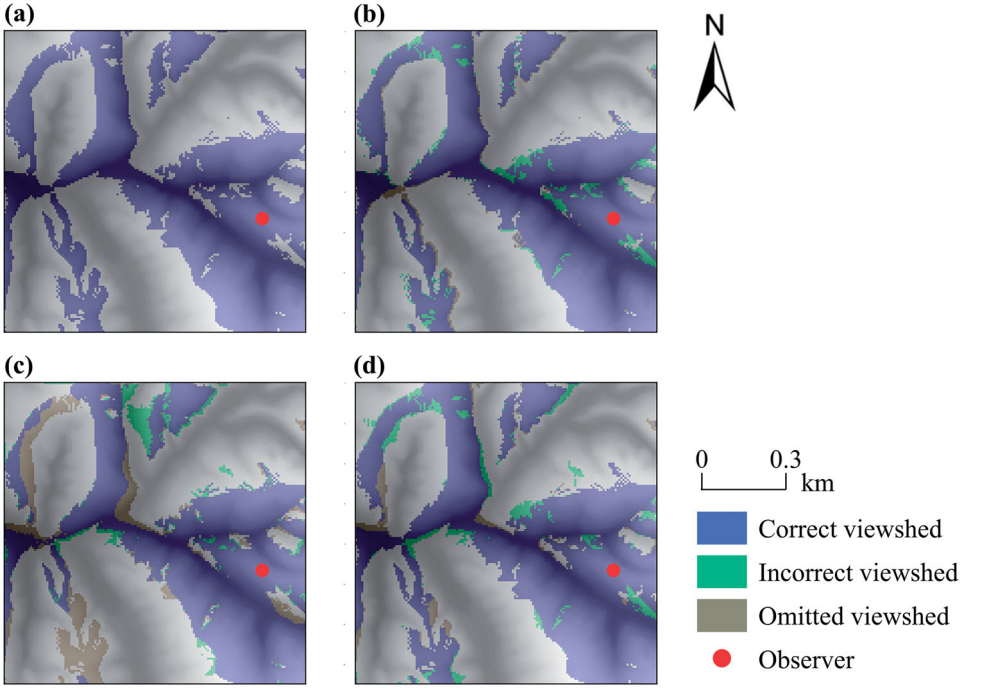


Figure 7. Comparison of results of different methods: (a) R3; (b) R2; (c) XDraw; (d) View-tree method.

Table 3. Error metrics for several methods.

Algorithm	R2		XDraw		View-tree method	
Metrics	η_{er} (%)	η_{or} (%)	η_{er} (%)	η_{or} (%)	η_{er} (%)	η_{or} (%)
Observation point 1	0.07	2.18	0.35	6.98	0.14	2.99
Observation point 2	0.01	2.44	0.06	12.75	0.04	4.12
Observation point 3	0.02	1.86	0.07	5.81	0.02	2.90
Observation point 4	2.93	3.90	9.27	17.23	3.71	8.79
Observation point 5	5.18	2.91	13.59	16.83	4.20	9.52
Observation point 6	5.52	3.03	9.23	17.67	4.97	7.70
Observation point 7	1.23	2.42	3.13	19.57	1.41	12.17
Observation point 8	4.40	1.66	8.84	13.46	3.58	9.80
Observation point 9	4.74	1.94	8.28	23.03	4.25	9.76
Observation point 10	3.85	2.81	5.80	21.96	2.84	9.12
Observation point 11	3.89	1.50	10.54	22.86	5.07	7.81
Observation point 12	4.66	3.27	8.67	21.75	4.34	9.92
Observation point 13	2.82	1.97	7.86	13.24	3.56	11.16
Observation point 14	1.91	2.60	10.85	22.05	2.38	10.98
Observation point 15	0.78	5.78	1.85	20.95	1.26	10.09
Average	2.80	2.68	6.56	17.07	2.78	8.45

4. Results

4.1. Precision of the calculated viewsheds

Figure 7 shows the operating precision of the different methods assessed. It can be observed that the results of view-tree method are similar to those of R2 and R3, which are high-precision methods based on LOS. The operating results of XDraw are much more different from those of R3, and the precision of XDraw is relatively lower. The error and distortion of XDraw will accumulate from the observation location layer by layer outward, with the result that this difference is especially obvious at far from the observation location.

As mentioned in Section 3.2, we used different terrain feature points as observation points and computed the viewshed as well as the precision separately. The results are shown in Table 3. It can be observed that view-tree method has the lowest error rate (2.78%) among the three methods, and its omission rate (8.45%) is significantly lower than that of XDraw (17.07%) and is still higher than that of R2 (2.68%) to some extent. This indicates that view-tree method has acceptable performance in terms of precision. Compared to R2, the view-tree method does best in not mistaking invisible cells for visible cells but has a slightly higher risk of mistaking visible cells for invisible cells. Additionally, although it is considered the most efficient method, XDraw has the highest error rate (6.56%) and omission rate (17.07%). Thus, its precision is relatively poor compared with that of the other two methods.

4.2. Efficiency comparison between different methods

The view-tree method, R2 and XDraw are substantially more efficient than R3 (Figure 8). Despite having the highest precision, R3 exhibits a relatively low efficiency. The time complexity of R3 is theoretically $O(N^3)$ while those for the view-tree method, R2 and XDraw are all theoretically $O(N^2)$. However, the difference in

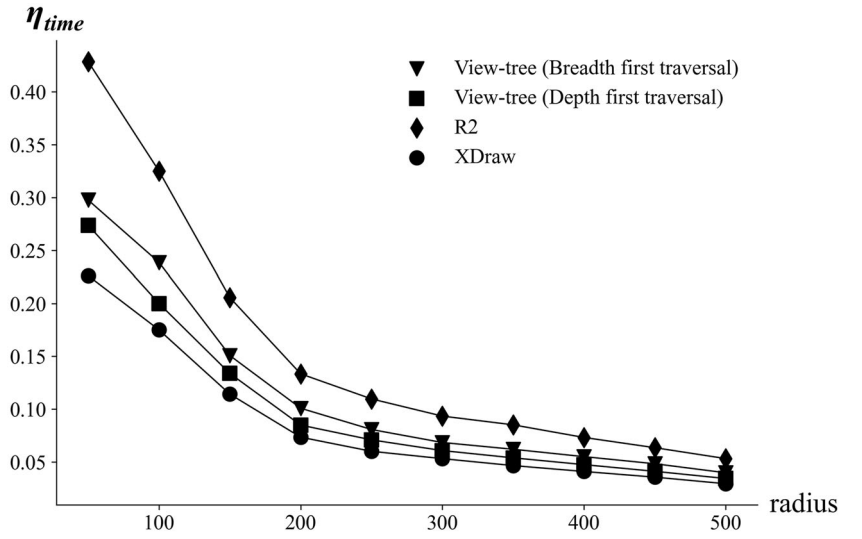


Figure 8. Comparison of operational efficiency of different methods.

characteristics of these methods leads to a difference in the efficiency of the actual experiments. Overall, XDraw has the highest relative operating efficiency. The view-tree method with depth-first traversal has the second-highest, costing 10% more time than XDraw, followed by the view-tree method with breadth-first traversal, costing 19% more time than XDraw. R2 is in the last place among the three methods in terms of efficiency, and the computing time of R2 is 1.8 times as long as that of XDraw.

The results here and in Section 4.1 indicate that XDraw achieves the highest efficiency, which is, however, based on a large sacrifice of precision. The view-tree method has an efficiency similar to that of XDraw but with a substantial improvement of precision, while R2 is relatively less efficient than view-tree method. R3 is the least efficient method among all methods, even though its precision is the highest. This difference tends to increase as the regional radius increases.

4.3. Sensitivity analysis

The experimental results are shown in Figures 9 and 10. Figure 9 shows it is clear that view-tree method provides different results under different R values, where L is the cell size. When $R \leq 0.40\,L$, the experimental results produce abrupt changes that coincide with the case of the too-small R mentioned in Section 2.3. That is, the view-tree only records cells in the due east, due west, due south, and due north directions of the observation location because only the cells that are traversed can be added to the tree. When R lies around $0.50\,L$, the precision of view-tree method is relatively higher, and its operating results are closer to the operating results of R3. As the value of R increases, the operating results of view-tree method will gradually change. When $R \geq 0.70\,L$, the results of view-tree method remain stable, which is similar to the operating results of XDraw.

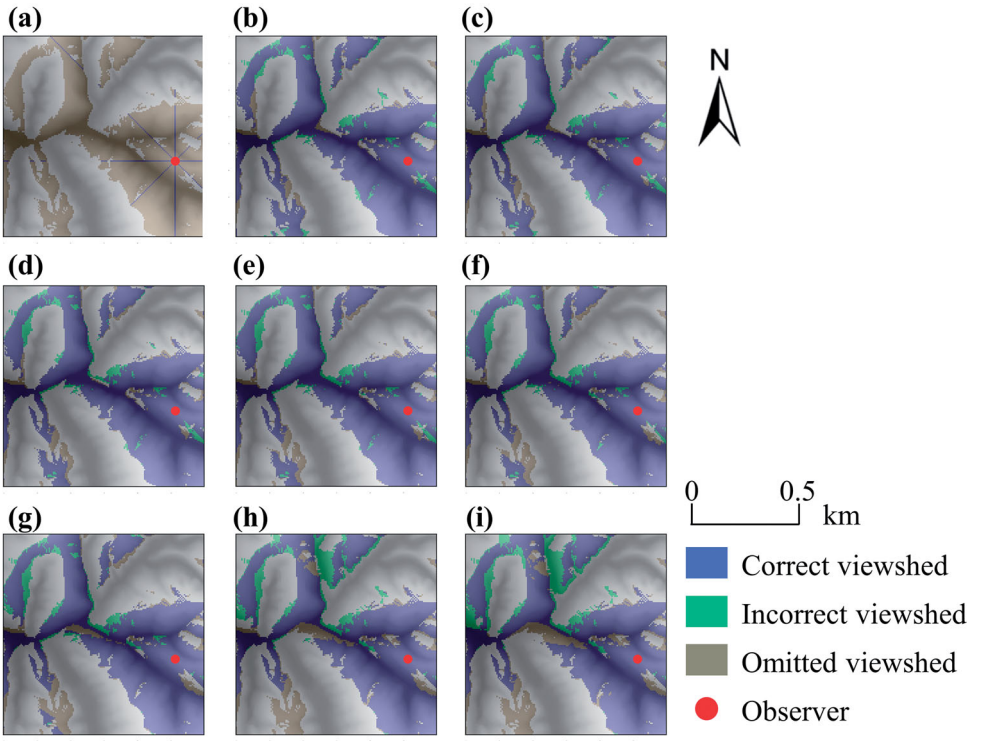


Figure 9. Operating results of view-tree method change with the value of R: the cell size is L: (a) $R = 0.40\,L$; (b) $R = 0.45\,L$; (c) $R = 0.50\,L$; (d) $R = 0.55\,L$; (e) $R = 0.60\,L$; (f) $R = 0.65\,L$; (g) $R = 0.70\,L$; (h) $R = 0.75\,L$; (i) $R = 0.80\,L$.

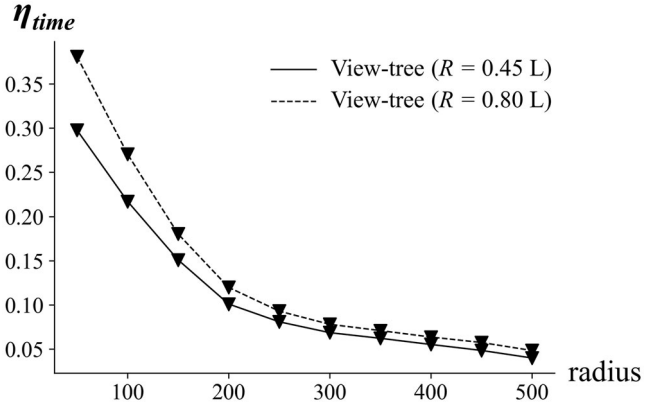


Figure 10. Operating efficiencies of view-tree method change with the value of R.

Figure 10 illustrates the operating efficiencies under different values of the parameter R. When $R \leq 0.40\,L$, the results are distorted and meaningless (Figure 9), thus the experiment on the operating efficiencies only focuses on cases where $R \geq 0.45\,L$. The effect of R on efficiencies has a certain regularity. Increased values of R require more time to construct the view-tree and compute the viewshed, and thus η_{time} decreases.

Compared with the η_{time} when $R = 0.45\,L$, the η_{time} when $R = 0.80\,L$ increases by 19% on average when breadth-first traversal is used. This range of variation brought by R is relatively modest compared with the changes among different methods.

Importantly, the results of the experiments reflect how the view-tree method relates to several current methods in principle. When the value of R is small, the judgment of view-block relationship is stricter, and the path traveled from the node of the view-tree up to the root node (i.e. the observer) is closer to a straight line. Therefore, the method has a high similarity in results with the current LOS-based methods. When R is larger, view-blocks will become common, and a single cell can use more elevation information from its neighboring cells to judge whether it is visible or not. This property is more similar to XDraw and other methods that use interpolation of the adjacent cells, so their results also tend to be similar. In summary, view-tree method brings together the characteristics of the two mainstream viewshed computation strategies, as reflected by the different R values.

5. Discussion

5.1. Advancement of view-tree method in terms of data structure

Before this method was proposed, methods for viewshed computation based on DEMs were mostly limited to the traditional data structure of regular square grid in terms of calculation and traversal, although each had its own characteristics and advantages.

On a 2-D plane, the occlusion relationships between observers and targets are very similar to those in the polar coordinate system with the observer as the pole and relationships to targets defined by azimuth and distance. However, regular square grid DEMs are usually described by the rectangular coordinate system and we cannot calculate the occlusion relationships directly. This is a major limitation in visibility analysis (Wu *et al.* 2021).

Some scholars noticed this problem and proposed the use of the line-rasterization method and interpolation method to abstract the LOS (Huss and Pumar 1997). This makes the regular square grid DEM adapted to the spatial relationships between locations in visibility analysis and successfully improves the precision of computation. Other scholars abstracted the distance relationships in the polar coordinate as the inner and outer relationships in 2-D array. DEMs were traversed from the inner to the outer layers from the observation location with the help of dynamic programming to improve the efficiency of computation. Methods such as XDraw were used in this process (Franklin and Ray 1994). Although the methods mentioned above have resolved the limitations of regular square grid DEM in viewshed computation to varying degrees, their ideas and processes have not departed from the grid structure. Several studies adapted the approach of transforming data structure and achieved ideal progress compared with traditional methods based on grid DEMs. Wu et al. (2021) proposed PDE coordinate system which can record the linear relationship of proximity, direction and elevation. Sanchez-Fernandez *et al.* (2021) proposed a methodology named skewed digital elevation model and improved the locality of memory accesses to speed-up viewshed computation.

In this study, the view-block relationship among cells in the regular square grid DEM is defined, and the corresponding tree data structure, named view-tree, is constructed with this relationship. Attributes such as hierarchy and depth and the organizational relationships between nodes in the tree data structure are well suited to describe spatial relationships and are commonplace for geographical problems, such as spatial indexes (Guttman 1984, Sellis *et al.* 1987) and spatial data compression (Samet 1982). Using the traversal algorithms of tree data structure, such as depth-first traversal, and breadth-first traversal, the traversal of view-tree and the judgment of each node's visibility are completed, and the viewshed is obtained as well. Therefore, the method for viewshed computation is advanced compared to the existing methods in its data structure. Its comprehensive performance is also superior generally in precision and efficiency.

Table 4. Error metrics for several methods for different terrain types.

Algorithm	R2		XDraw		View-tree method	
Metrics	η_{er} (%)	η_{or} (%)	η_{er} (%)	η_{or} (%)	η_{er} (%)	η_{or} (%)
Loess hilly-gully region	2.80	2.68	6.56	17.07	2.78	8.45
Mountainous region	4.12	2.59	6.19	24.02	4.46	7.34
Loess ridge-gully region	5.91	3.18	11.59	19.64	4.43	9.12
Average	4.28	2.81	8.11	20.24	3.88	8.30

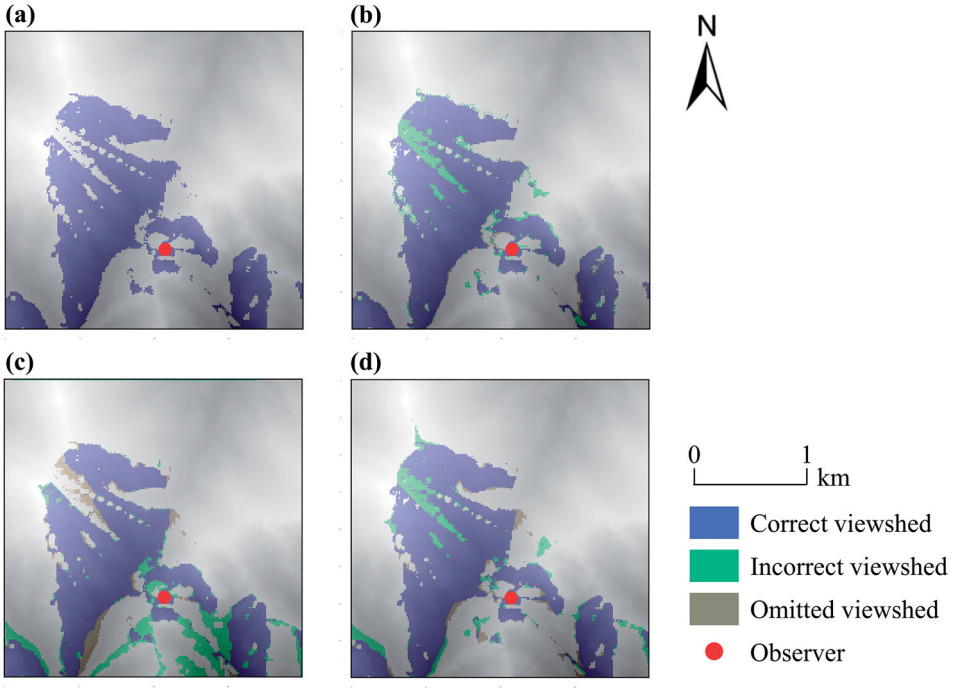


Figure 11. Comparison of results of different methods in a typical bedrock mountainous region with a regional average altitude of 2146 m in Qinling Mountains, Shanxi, China: (a) R3; (b) R2; (c) XDraw; (d) view-tree method.

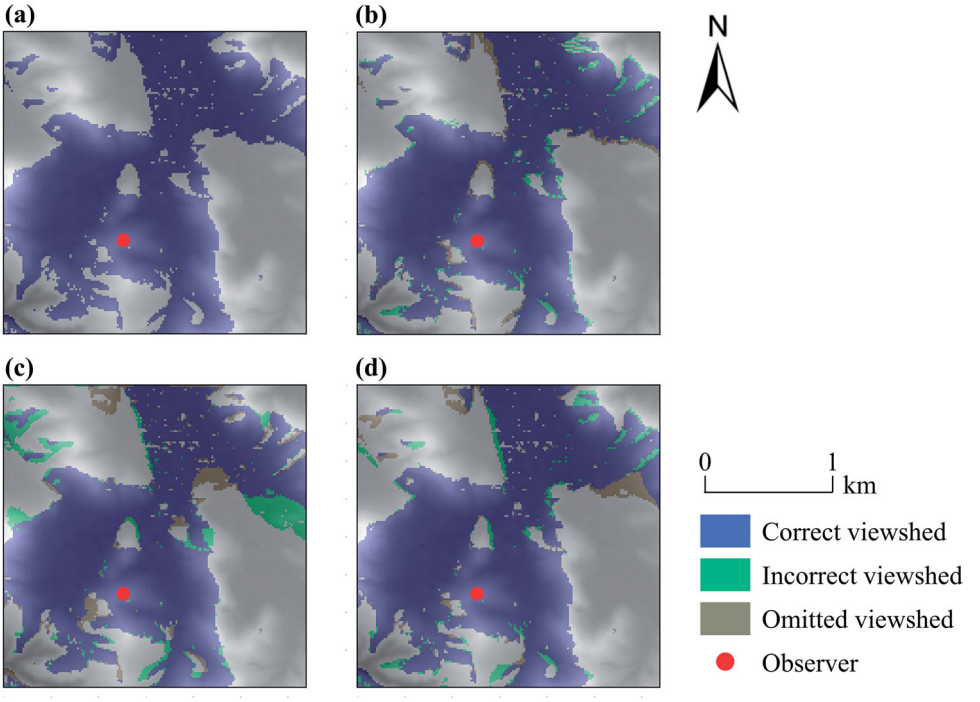


Figure 12. Comparison of results of different methods in a typical loess ridge-gully region with a regional average altitude of 1512 m in Yan'an City, Shanxi, China: (a) R3; (b) R2; (c) XDraw; (d) view-tree method.

5.2. Validation of view-tree method for different terrain types

Viewshed has different characteristics for different landforms and terrain types (Lv *et al.* 2017). In this section, we added two other landform regions (i.e. bedrock mountainous terrain and loess ridge-gully terrain) to validate the precision of view-tree method for regions that are geomorphologically different. The resampled resolution of all experimental data is 2 m.

Table 4 presents the results of the error rate and omission rate of R3, R2, XDraw and view-tree method in all terrain types involved. For loess hilly-gully terrain, as is mentioned in Section 4.1, the view-tree method has the lowest error rate (2.78%) while R2 has the lowest omission rate (2.68%). For bedrock mountainous terrain, R2 performs best both in error rate (4.12%) and in omission rate (2.59%) because this terrain type possesses high continuity. The view-tree method also performs well here with an error rate of 4.46% and an omission rate of 7.34%. For loess ridge-gully terrain, R2 performs slightly worse than it does for the other two terrain types but still has the lowest omission rate (3.18%) and view-tree method performs best in error rate (4.43%).

The operating results of R3, R2, XDraw and the view-tree method are shown in Figures 11 and 12. The outputs of view-tree method have evident similarity with those of R3 in both viewshed area and spatial distribution. However, XDraw has a lower precision compared with other methods and has obvious errors for all experimental regions. The experimental results above indicate that the view-tree method performs stably and can retain its precision for a range of terrain types.

6. Conclusion

We introduce a new tree data structure, view-tree, to transform a DEM and propose a new method to compute viewsheds based on the view-tree. The method overcomes the limitations of regular square grid DEMs in traditional viewshed computation methods. Moreover, it is superior in precision and efficiency metrics. The method is also an important supplement in transforming data structure to solve viewshed computation problems.

The study obtained the following conclusions through a series of experiments. The efficiency of the view-tree method is 40% faster than R2. The highest efficiency is achieved when depth-first traversal is used. The precision of the view-tree method is guaranteed on the premise of ensured efficiency. Using nearly the same time, the view-tree method reduces the error rate by approximately 60% and the omission rate by 50% compared with XDraw. The operating result of the view-tree method is very similar to that of R3, with an error rate of 2.78% and an omission rate of 8.45%. The R variable in the view-tree method has a considerable impact on the judgment of the view-block relationship and the operating results of the method. This method has the highest precision when R is set to approximately half of the cell size.

The method has the following problems. As is shown in Sections 4.1 and 5.2, the view-tree method performs worse than R2 in the case of omission rate. The reason for this is that each node in a view-tree can only acquire information from the parent node and the LOS calculated by the view-tree method is therefore still inferior to that by R2 in terms of quality. The strategies for improving the quality of LOS in the view-tree method will be a focus of future work. The view-tree method is currently implemented as a serial computation method. Implementing it under a parallel framework is an important objective in the next stage of research.

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No potential conflict of interest was reported by the author(s).

Data and codes availability statement

The data and codes that support this study are available in https://doi.org/10.6084/m9.figshare.16809802.v2.

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